Databricks Interview Questions & Answers (0–5 Years Experience)

1. What is the Databricks Lakehouse Architecture?

Answer:
Databricks Lakehouse combines the best of data lakes (low-cost, scalable storage like ADLS/S3) and data warehouses (ACID transactions, schema enforcement, BI performance).
It is built on the **Delta Lake** format which brings reliability to data lakes.

Key features:

- ACID transactions
- Schema enforcement & evolution
- DML operations (MERGE/UPDATE/DELETE)
- Time Travel
- High-performance caching

2. Scenario: You have late-arriving data every day. How do you handle it using Databricks?

Answer:
Use **MERGE INTO** with **Delta Lake** to upsert.

Code:

```
MERGE INTO target t
USING source s
ON t.id = s.id
WHEN MATCHED THEN UPDATE SET *
WHEN NOT MATCHED THEN INSERT *
```

Delta Lake ensures ACID compliance and avoids duplicates.

3. What is Autoloader? When do you use it?

Answer:
Autoloader is a high-performance ingestion feature in Databricks used for **incremental file ingestion** from cloud storage.

Why use it?

- Automatically detects new files
- Supports schema inference & evolution
- Scales to billions of files

- Works best for streaming ingestion

Example:

```
df = spark.readStream.format("cloudFiles")\
  .option("cloudFiles.format", "json")\
  .load("/mnt/raw/")
```

4. Scenario: You have schema changes in your raw data. How do you manage this in Databricks?

Answer:
Use **Schema Evolution** with Delta Lake.

Example:

```
ALTER TABLE users SET TBLPROPERTIES (
  'delta.schema.autoMerge.enabled' = 'true'
);
```

5. How does Delta Lake support Time Travel?

Answer:
You can query older versions of data using **VERSION AS OF** or **TIMESTAMP AS OF**.

```
SELECT * FROM customers VERSION AS OF 3;
```

Useful for audit, rollback, debugging.

6. What is the difference between Managed and External Tables in Databricks?

Answer:

Managed Table	External Table
Databricks manages metadata + data	Only metadata is managed
Data is stored in DBFS	Data stored in ADLS/S3 path
Dropping table removes data	Dropping table keeps data

7. What is Delta Live Tables (DLT)? Why is it used?

Answer:
DLT is a framework to build **reliable & declarative ETL pipelines**.

Features:

- CDC support
- Auto-recovery
- Quality rules (EXPECT statements)
- Lineage tracking

Example:

```
@dlt.table
def cleaned_orders():
    return dlt.read("raw_orders").where("amount > 0")
```

8. Scenario: How do you implement Change Data Capture (CDC) in Databricks?

Answer:
Use **Delta MERGE + SCD Type 2 design**.

```
MERGE INTO target t
USING changes c
ON t.id = c.id
WHEN MATCHED AND c.is_deleted = true THEN DELETE
WHEN MATCHED THEN UPDATE SET ...
WHEN NOT MATCHED THEN INSERT ...
```

9. Difference between `display()`, `show()`, and `collect()`?

Answer:

Function	Purpose
`show()`	Displays first rows of DataFrame
`display()`	Databricks UI formatting + visualization
`collect()`	Moves entire DataFrame to driver (dangerous for large data)

10. What is Z-Ordering in Databricks?

Answer:
Optimizes data skipping by co-locating related data files.

```
OPTIMIZE sales ZORDER BY (customer_id);
```

Makes queries faster.

11. Scenario: A table query is extremely slow. What do you do?

Answer:

- Use **OPTIMIZE + ZORDER**
- Check skewness using `df.explain()`
- Repartition based on high-cardinality column

- Use Delta Caching
- Reduce shuffle operations
- Convert Parquet → Delta

12. What is Photon Engine?

Answer:
Next-gen query execution engine for Databricks.
Optimized for:

- Vectorized SQL processing
- Fast Delta I/O
- Low-latency analytics

Used automatically in SQL Warehouse.

13. How do you handle data skew in Databricks?

Answer:

- Use `salting` (adding random value to key)
- `repartition()`
- `broadcast()` small table
- Use Adaptive Query Execution (AQE)

```
spark.conf.set("spark.sql.adaptive.enabled", "true")
```

14. Scenario: You must join a huge fact table with a tiny dimension. What optimization do you use?

Answer:
Use **Broadcast Join**.

```
df.join(broadcast(dimDf), "id")
```

Reduces shuffle.

15. Explain Medallion Architecture (Bronze → Silver → Gold).

Answer:

Layer	Purpose
Bronze	Raw data ingestion
Silver	Cleaned & normalized
Gold	Business aggregated, ready for analytics

Used widely in Databricks Data Engineering pipelines.

16. Scenario: Your pipeline fails daily due to bad records. How do you fix it?

Answer:
Enable **data quality rules** using DLT:

```
EXPECT amount > 0 ON VIOLATION DROP ROW;
```

17. What is a Cluster Policy?

Answer:
Controls cluster configuration for governance:

- enforce node type
- limit worker count
- restrict libraries
- cost control

18. What are SQL Warehouses?

Answer:
Dedicated compute for BI + Analytics.
Used for dashboards, PowerBI, Tableau.

19. What is MLflow in Databricks?

Answer:
Tracks ML experiments:

- parameters
- metrics
- artifacts
- model registry

Used to deploy models into production.

20. Explain Unity Catalog. Why is it important?

Answer:
Unified governance layer for Databricks.

Features:

- Central metadata management
- Fine-grained access control
- Lineage visualization
- Cross-workspace sharing